The above two means are in decimal places so for the simplicity we use this formula to compute b_{YX}.

$$b_{YX} = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}$$

$$= \frac{8\times 1856 - 63\times 220}{8\times 555 - 63\times 63}$$

$$= \frac{14848 - 13860}{4440 - 3969}$$

$$= \frac{988}{471}$$

$$b_{YX} = 2.098$$

The regression equation of Y on X,

$$\hat{Y} - \bar{y} = b_{YX}\left(x - \bar{x}\right)$$

$$\hat{Y} - 27.5 = 2.098\ (x - 7.875)$$

$$\hat{Y} - 27.5 = 2.098\ x - 16.52$$

$$\hat{Y} = 2.098x + 10.98$$

When $x = 15$,

$$\hat{Y} = 2.098 \times 15 + 10.98$$

$$\hat{Y} = 31.47 + 10.98$$

$$= 42.45$$

Regression equation of X on Y,

$$\hat{X} - \bar{x} = b_{XY}\left(y - \bar{y}\right)$$

$$b_{XY} = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} y_i^2 - \left(\sum_{i=1}^{n} y_i\right)^2}$$

$$= \frac{8\times 1856 - 63\times 220}{8\times 6780 - 220\times 220}$$

$$= \frac{14848 - 13860}{54240 - 48400}$$

$$= \frac{988}{5840}$$

$$b_{XY} = 0.169$$

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